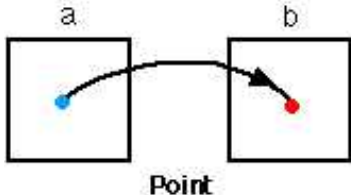
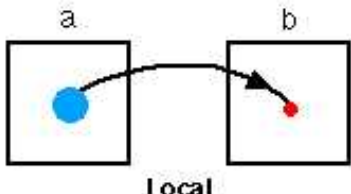
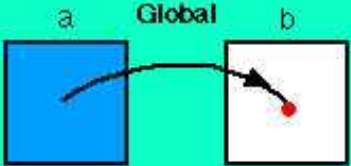


Επεξεργασία Εικόνας στο Πεδίο Συχνοτήτων

Τύποι Λειτουργιών

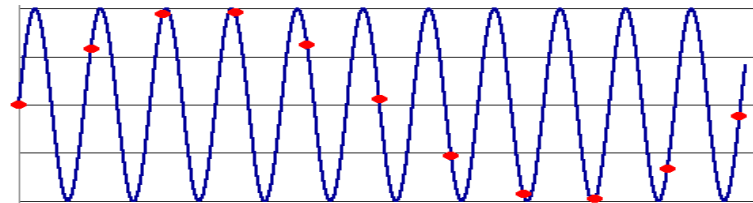
Οι λειτουργίες που μπορούν να εφαρμοσθούν σε ψηφιακές εικόνες για να μετασχηματίσουν την αρχική εικόνα $a[m,n]$ στην τελική εικόνα $b[m,n]$ (ή κάποια άλλη αναπαράσταση) ανήκουν σε 3 κατηγορίες:

Λειτουργία	Χαρακτηρισμός	Γραφική αναπαράσταση
σημειακή	η τελική τιμή στο (x, y) εξαρτάται μόνο από την αρχική τιμή στο (x, y) . Πολυπλοκότητα / pixel : <i>const</i>	
τοπική	η τελική τιμή στο (x, y) εξαρτάται από τις αρχικές τιμές μιας $P \times P$ γειτονιάς γύρω από το (x, y) . Πολυπλοκότητα / pixel : P^2	
καθολική	η τελική τιμή στο (x, y) εξαρτάται από όλες τις τιμές της αρχικής $N \times N$ εικόνας. Πολυπλοκότητα / pixel : N^2	

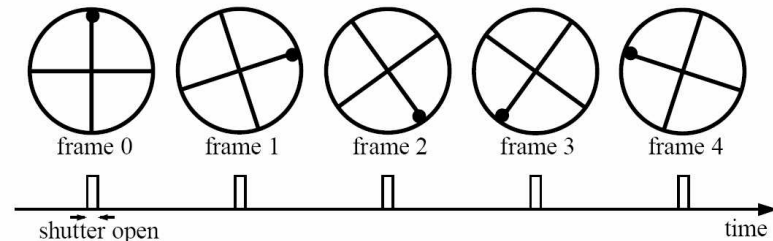
Aliasing

- occurs when the sampling rate is not high enough to capture the amount of detail in the signal/image
- Can result to wrong signal/image—an *alias*

- Signal alias



- Alias in video



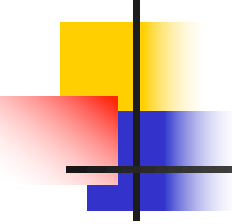
- To do sampling right, need to understand the signal/image structure

Image resampling can lead to aliasing



The jaggies!





Even worse for synthetic images



Jean Baptiste Fourier (1768-1830)

- had a crazy idea (1807):

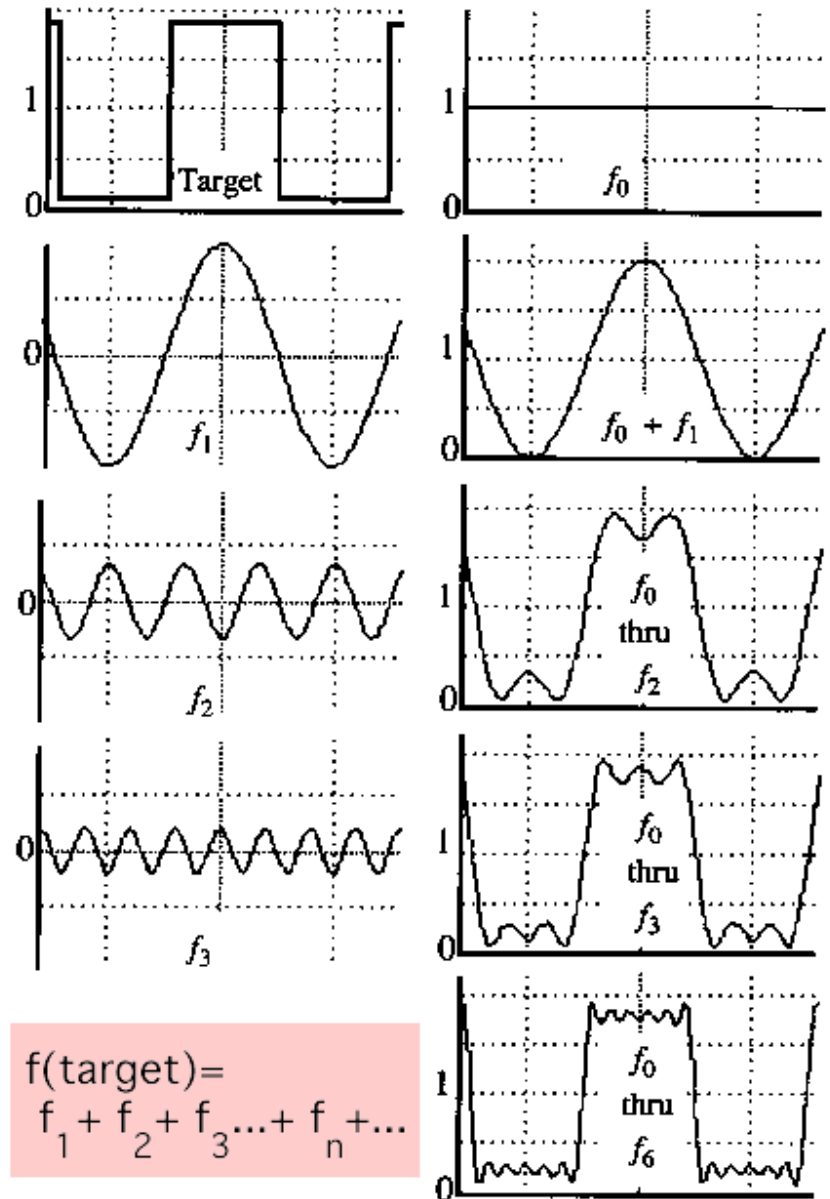
Any periodic function can be rewritten as a weighted sum of sines and cosines of different frequencies.

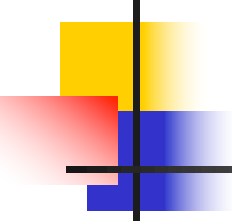
- Don't believe it?
 - Neither did Lagrange, Laplace, Poisson and other big wigs
 - Not translated into English until 1878!
- But it's true!
 - called **Fourier Series**



A sum of cosines

- Our building block:
 $A \cos(\omega x + \varphi)$
- Add enough of them to get any signal $f(x)$ you want!
- How many degrees of freedom?
- What does each control?
- Which one encodes the coarse vs. fine structure of the signal?



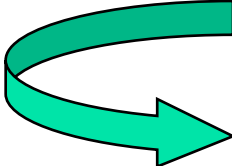


Fourier Transform

- We want to understand the frequency ω of our signal. So, let's reparametrize the signal by ω instead of x :



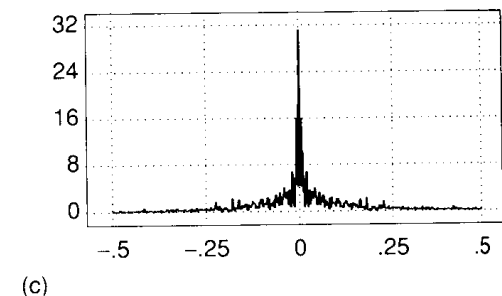
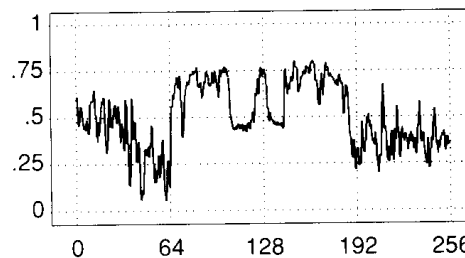
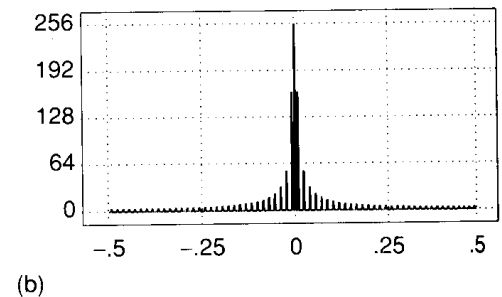
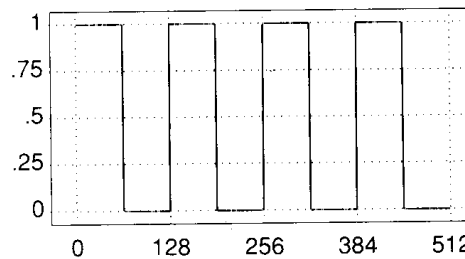
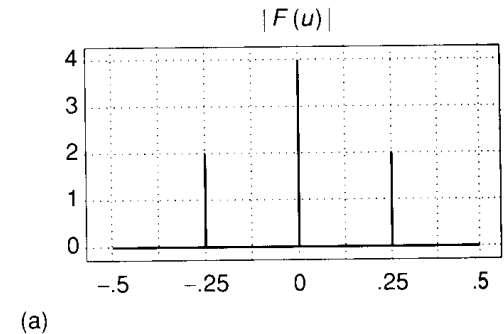
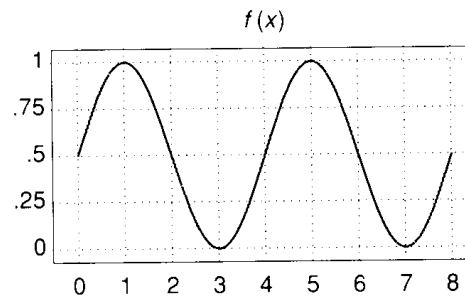
- For every ω from 0 to ∞ , $F(\omega)$ holds the amplitude A and phase ϕ of the corresponding sine
 - How can F hold both? Complex number trick!

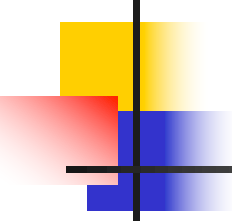

$$F(\omega) = R(\omega) + j I(\omega)$$

$$A = \sqrt{R^2(\omega) + I^2(\omega)}, \quad \phi = \tan^{-1} \frac{I(\omega)}{R(\omega)}$$

Frequency Spectra

Usually, amplitude is more interesting than phase.





Finally:

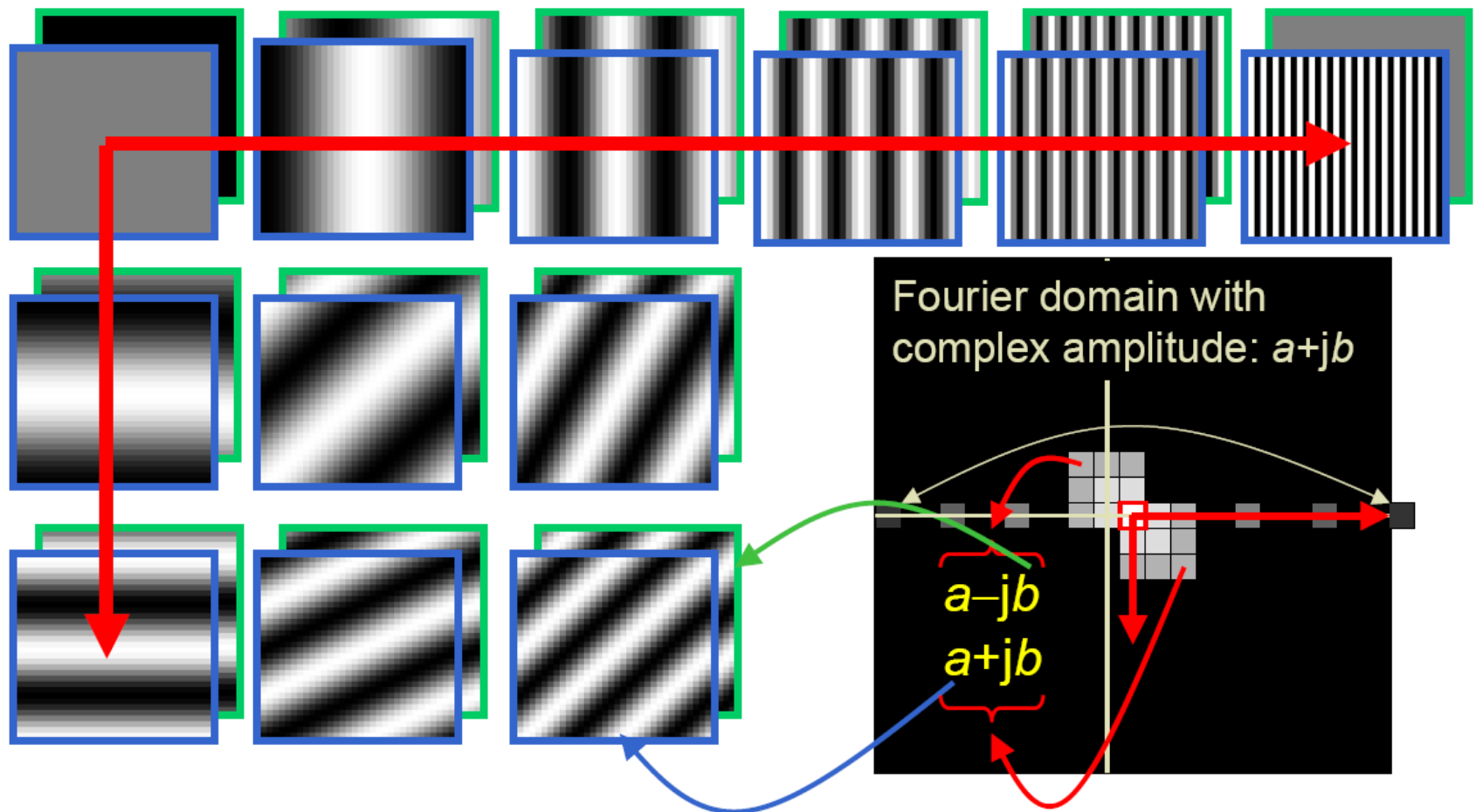
Fourier Transform :
$$F(\omega) = \int_{-\infty}^{+\infty} f(x)e^{-i\omega x} dx$$

Inverse Fourier Transform :
$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega)e^{i\omega x} d\omega$$

where, $e^{i\omega x} = \cos(\omega x) + i \sin(\omega x)$

is given by the well known Euler formula.

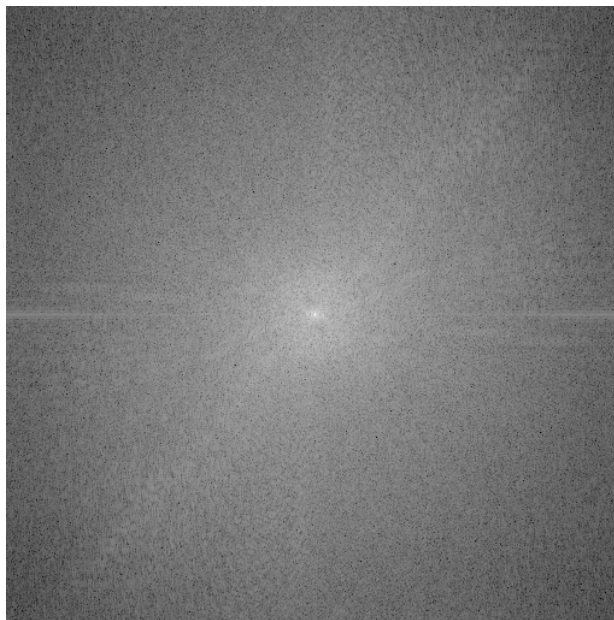
Extension to 2D



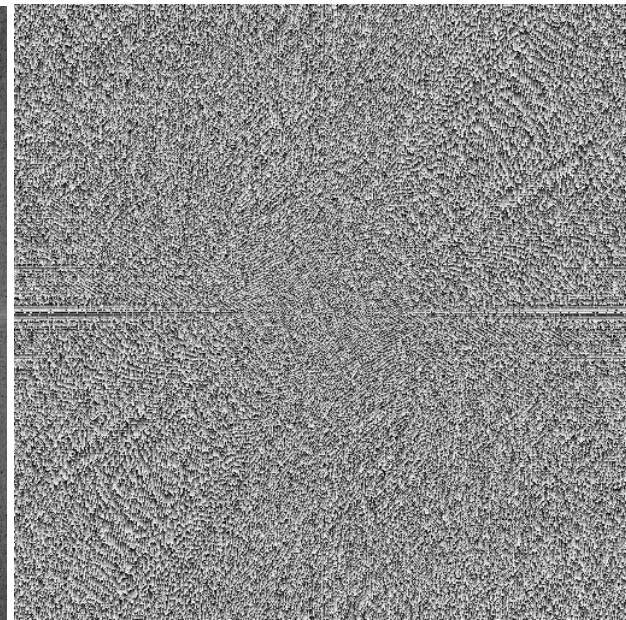
DFT of the cheetach image



The cheetach image



DFT magnitude

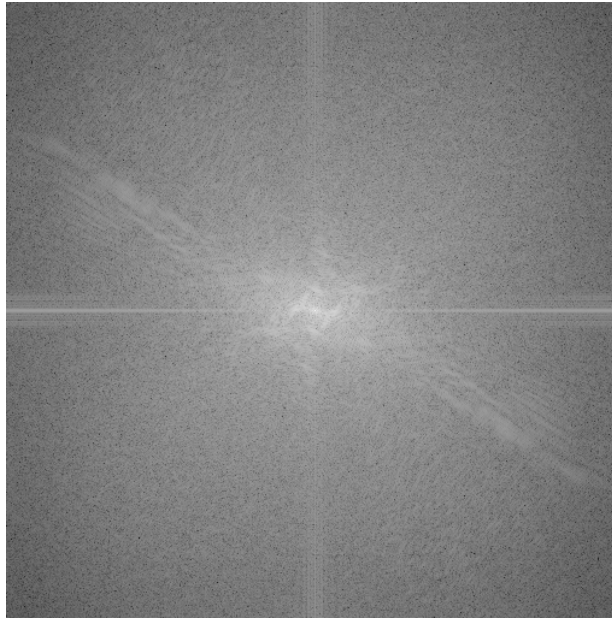


DFT phase

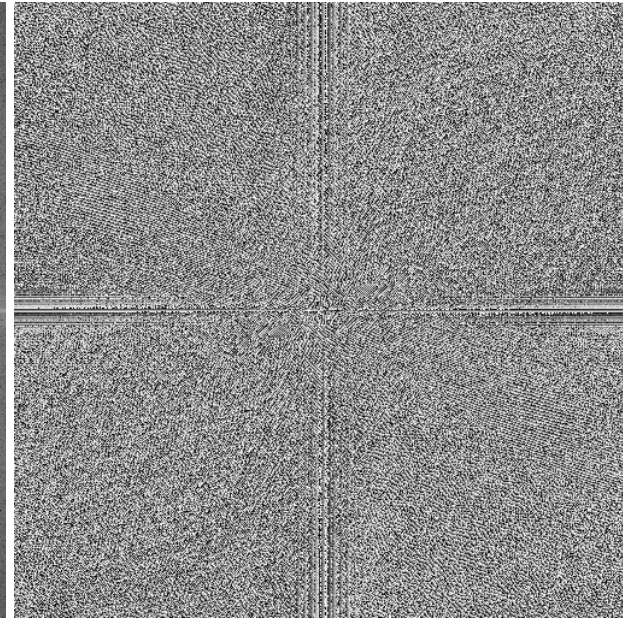
DFT of the zebra image



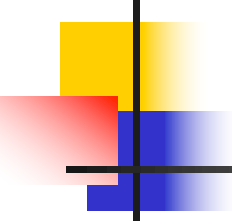
The zebra image



DFT magnitude

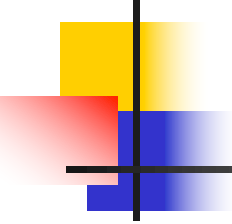


DFT phase

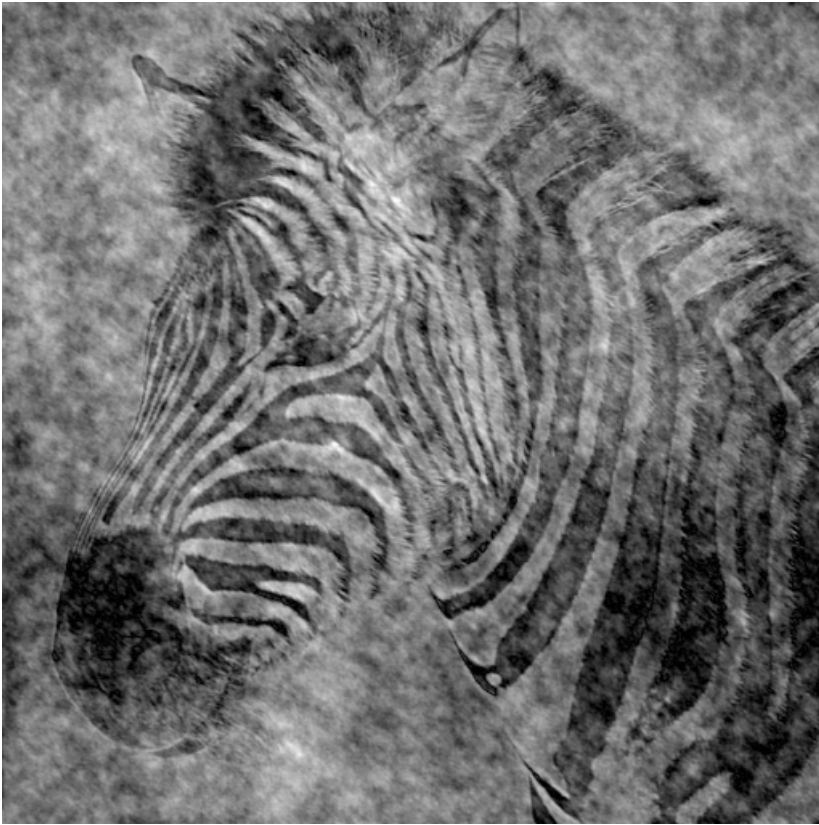


Curious things about FT on images

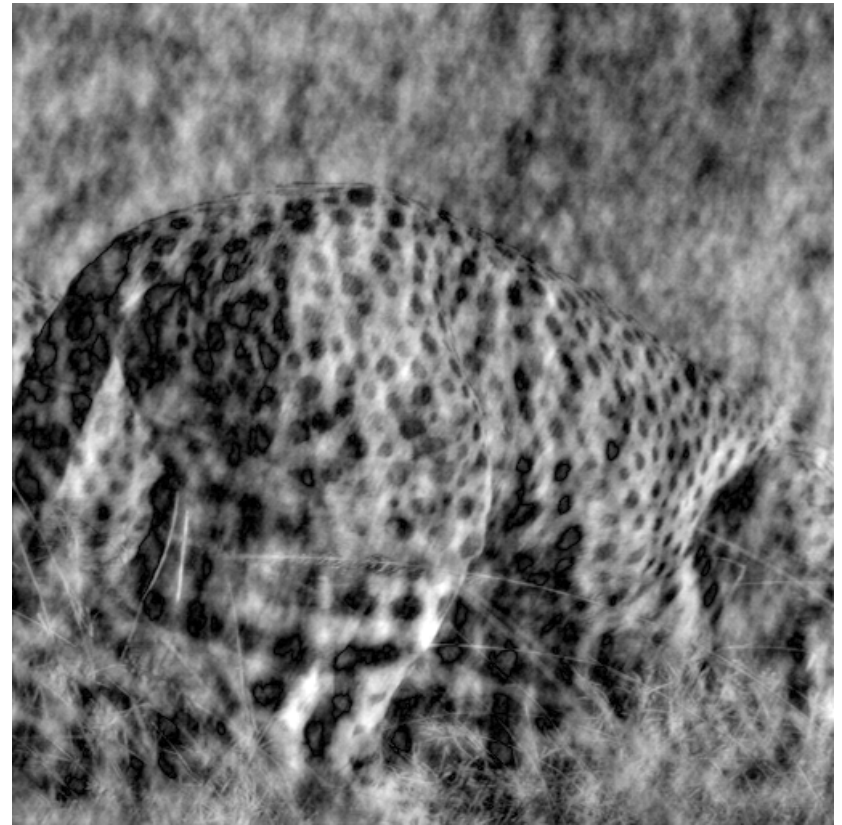
- The magnitude spectra of all natural images are quite similar
 - Large values on low-frequencies, falling off in high frequencies
- Most information in the image is carried in the phase, not the amplitude
 - Seems to be a fact of life
 - Not quite clear why



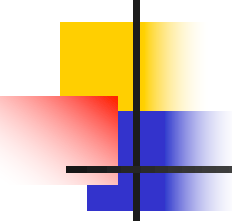
Reconstruction



zebra phase, cheetah magnitude



cheetah phase, zebra magnitude



Fourier transform properties

- The Fourier transform is linear
- There is an inverse FT
- The FT of a Gaussian is a Gaussian
- Need for fast implementations led to the Fast Fourier Transform (FFT) algorithm realized by DSP processors
- The convolution theorem:

The Fourier transform of the convolution (product) of two functions is the product (convolution) of their respective Fourier transforms

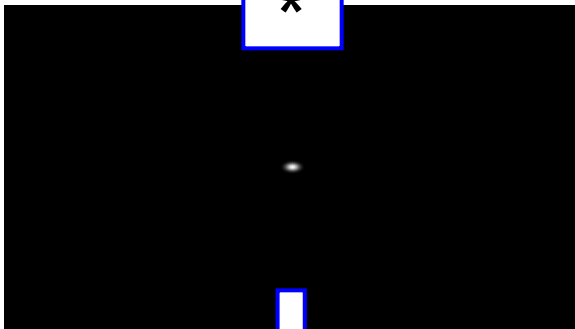
2D convolution theorem example

$f(x,y)$



*

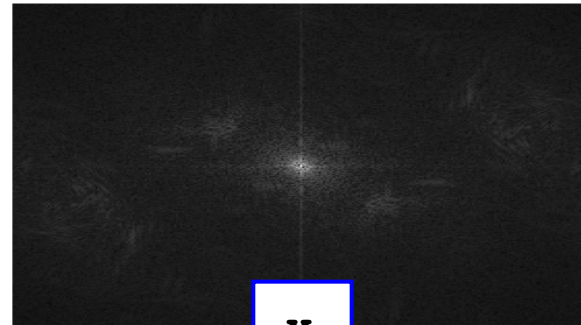
$h(x,y)$



$g(x,y)$

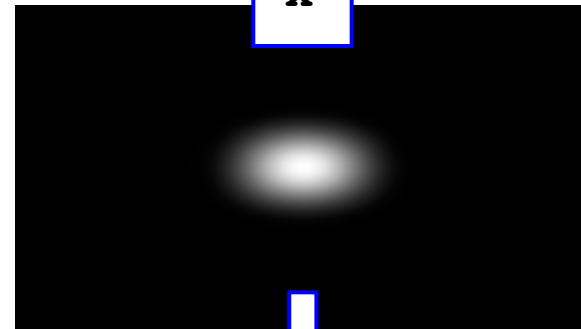


$|F(s_x,s_y)|$

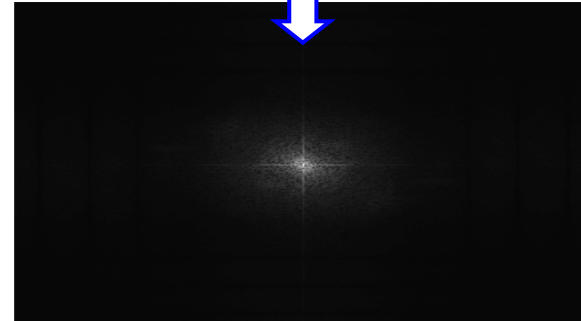


x

$|H(s_x,s_y)|$



$|G(s_x,s_y)|$



Filtering in the frequency domain

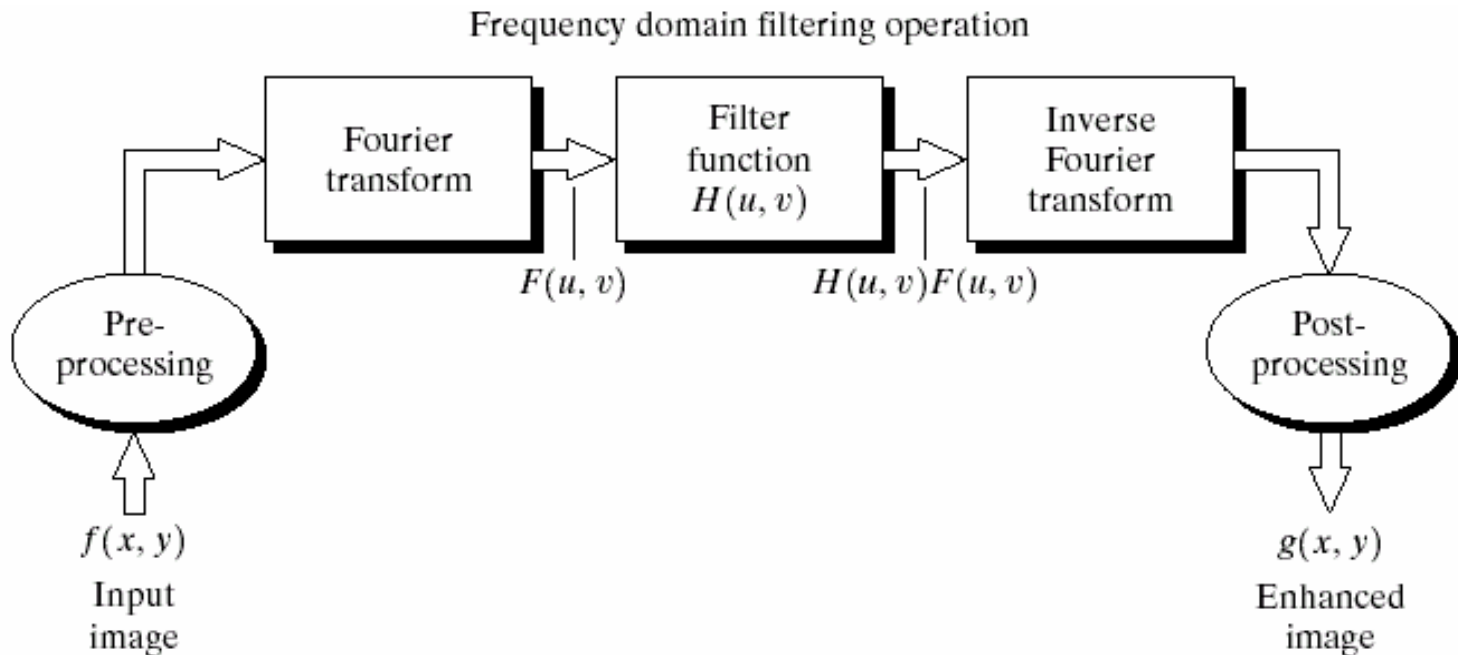
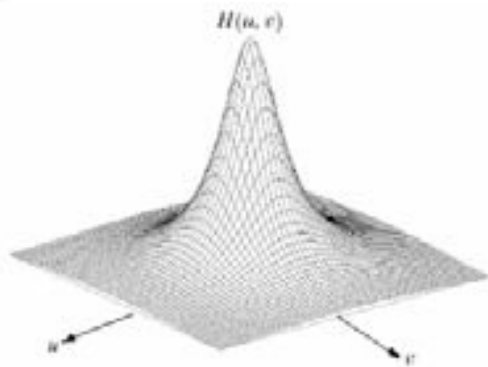
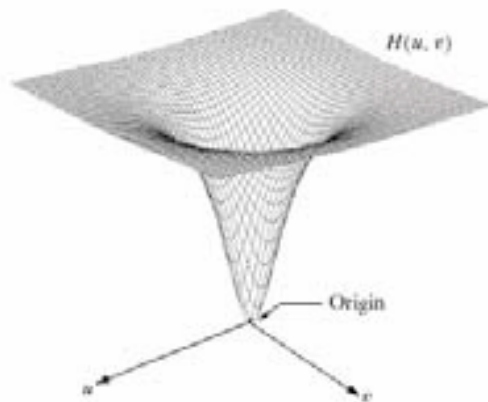


FIGURE 4.5 Basic steps for filtering in the frequency domain.



Low pass filter

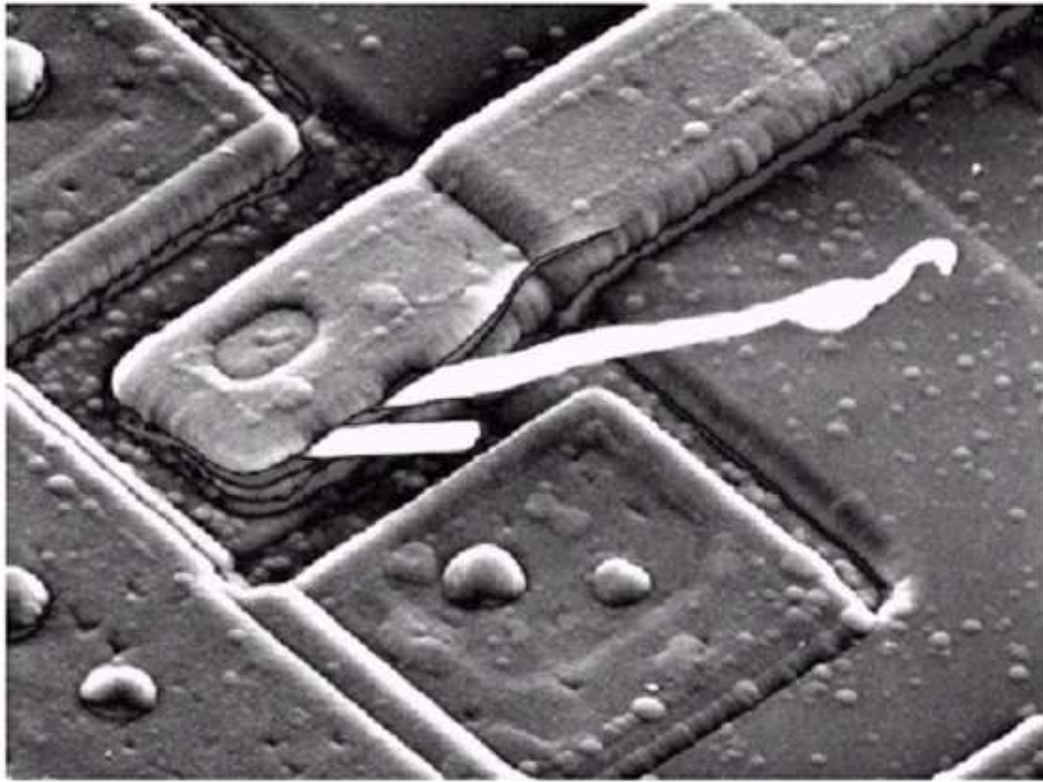


High pass filter

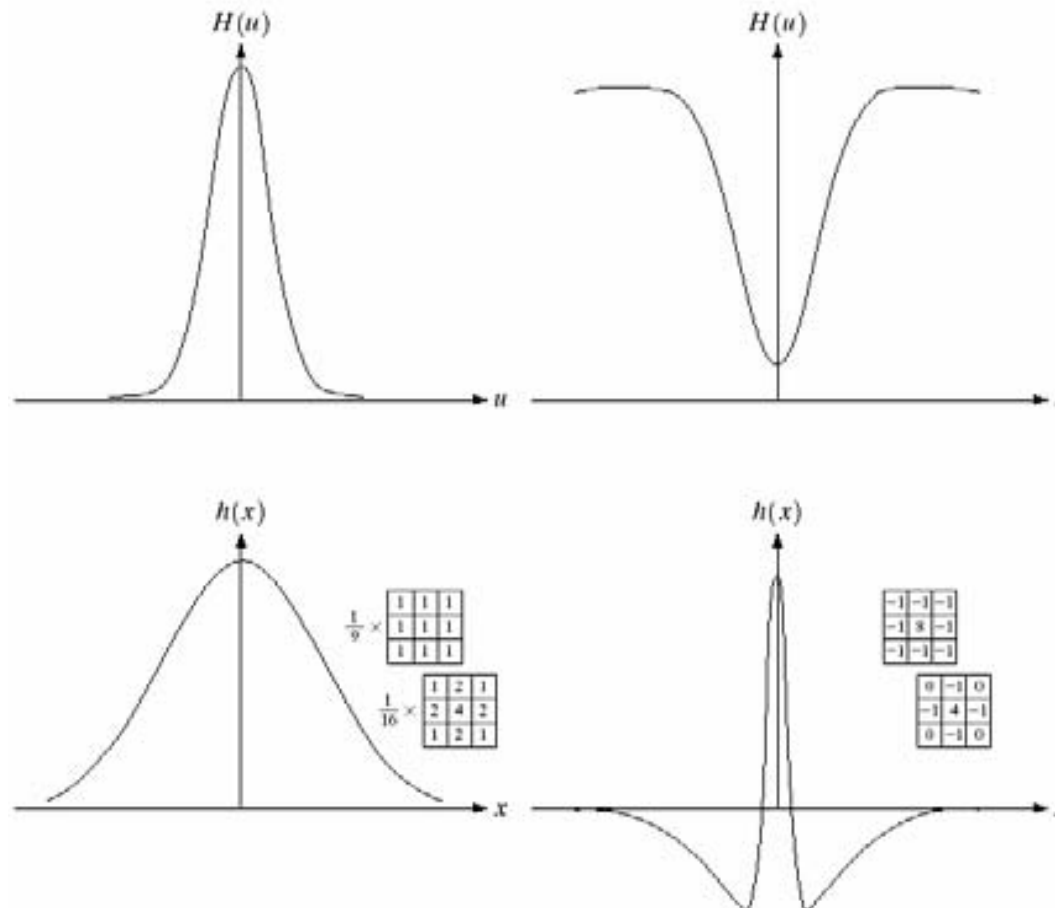
a b
c d

FIGURE 4.7 (a) A two-dimensional lowpass filter function. (b) Result of lowpass filtering the image in Fig. 4.4(a). (c) A two-dimensional highpass filter function. (d) Result of highpass filtering the image in Fig. 4.4(a).

Add the $\frac{1}{2}$ of filter height to $F(0,0)$
of the high pass filter



Correspondence between filters in spatial and frequency domains



a b
c d

FIGURE 4.9

(a) Gaussian frequency domain lowpass filter.

(b) Gaussian frequency domain highpass filter.

(c) Corresponding lowpass spatial filter.

(d) Corresponding highpass spatial filter. The masks shown are used in Chapter 3 for lowpass and highpass filtering.

The Ideal Lowpass filter: ILPF

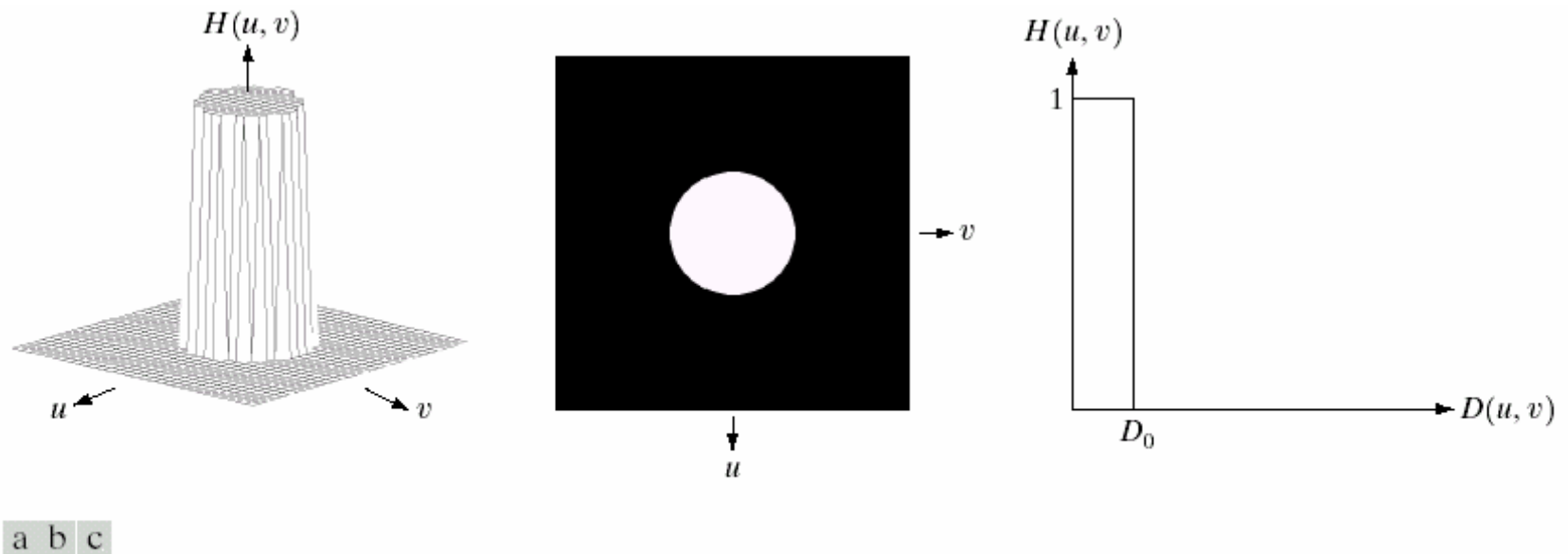
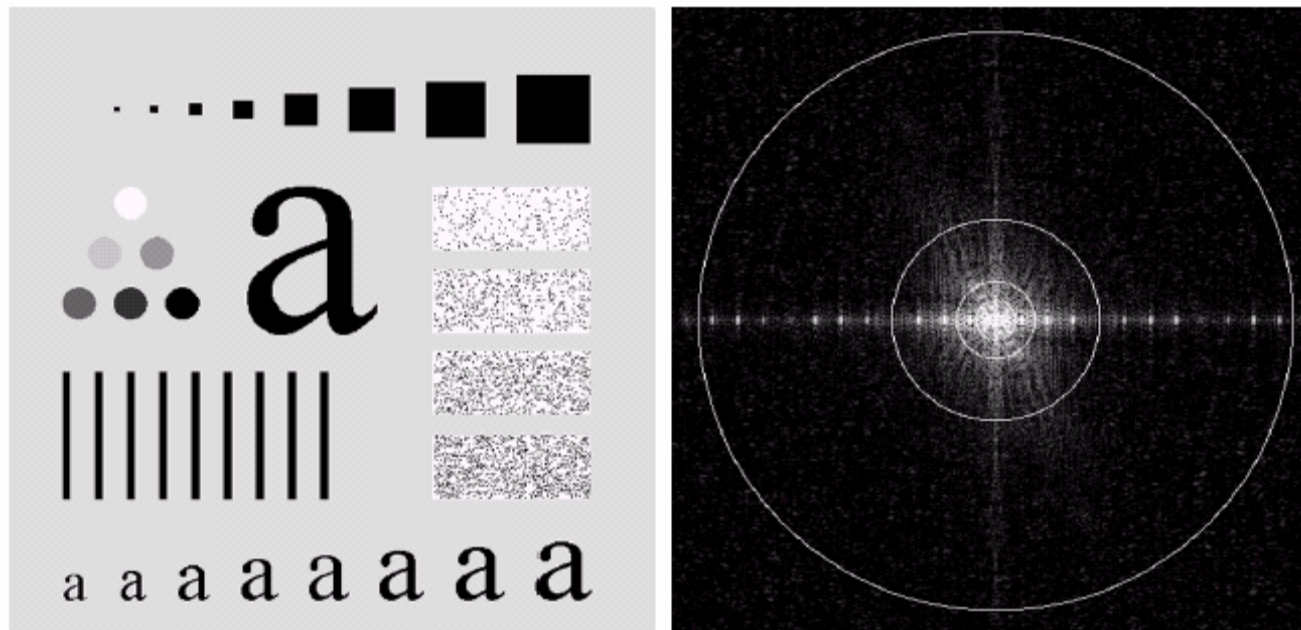


FIGURE 4.10 (a) Perspective plot of an ideal lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross section.

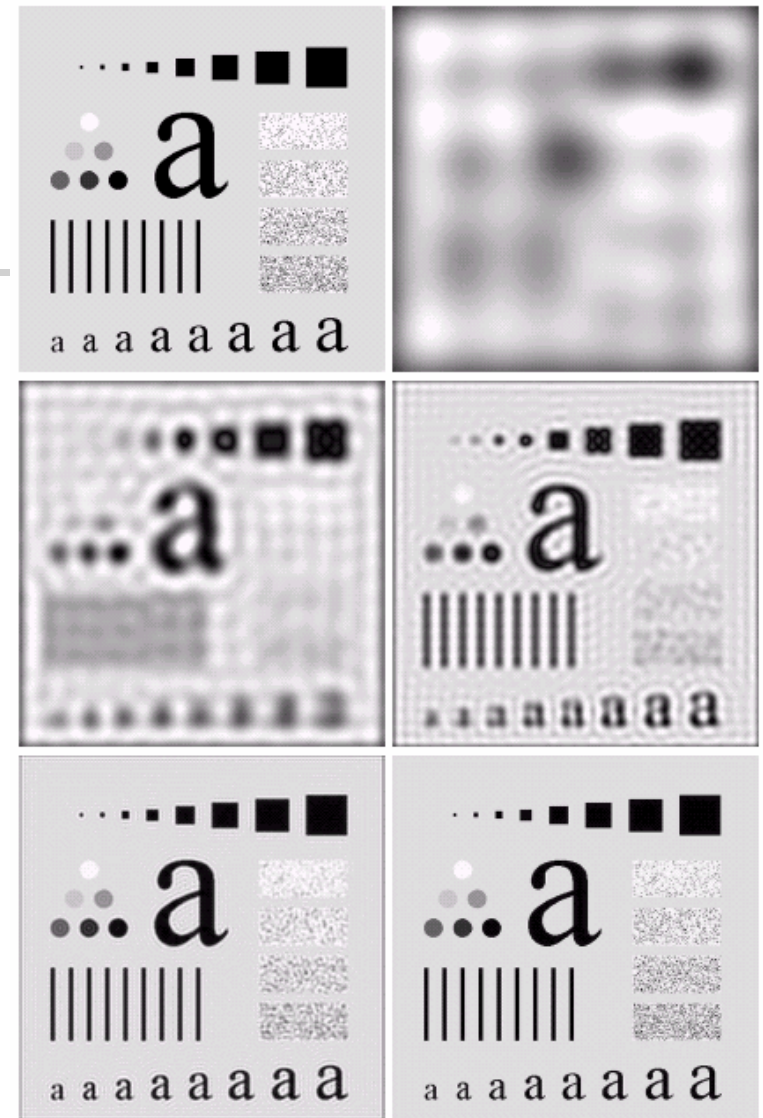
Image power circles



a b

FIGURE 4.11 (a) An image of size 500×500 pixels and (b) its Fourier spectrum. The superimposed circles have radii values of 5, 15, 30, 80, and 230, which enclose 92.0, 94.6, 96.4, 98.0, and 99.5% of the image power, respectively.

Result of ILPF



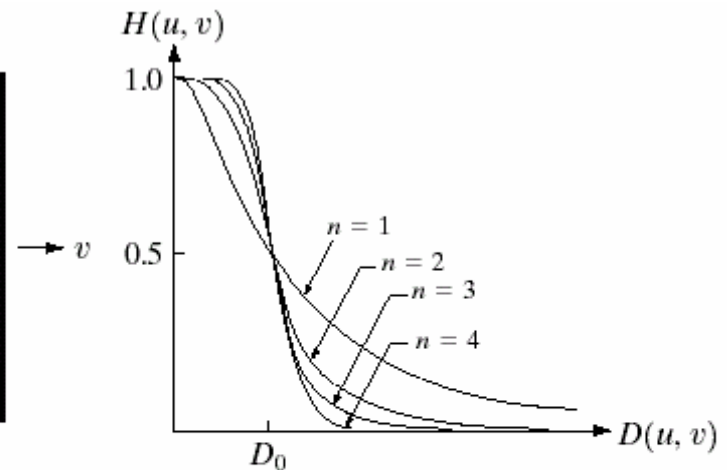
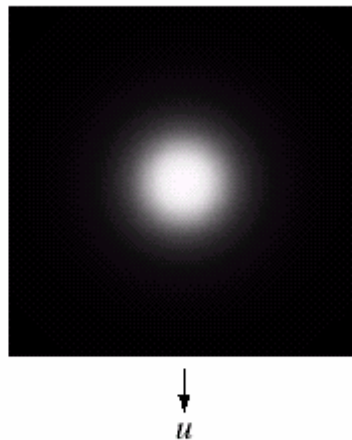
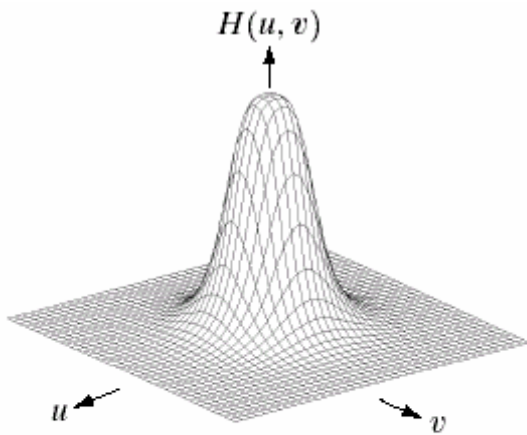
a	b
c	d
e	f

FIGURE 4.12 (a) Original image. (b)–(f) Results of ideal lowpass filtering with cutoff frequencies set at radii values of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). The power removed by these filters was 8, 5.4, 3.6, 2, and 0.5% of the total, respectively.

Butterworth Lowpass Filter: BLPF

The Butterworth filter transfer function:

$$H(u, v) = \frac{1}{1 + [D(u, v)/D_0]^{2n}}$$



a b c

FIGURE 4.14 (a) Perspective plot of a Butterworth lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.

Example BLPF

BLPF

ILPF

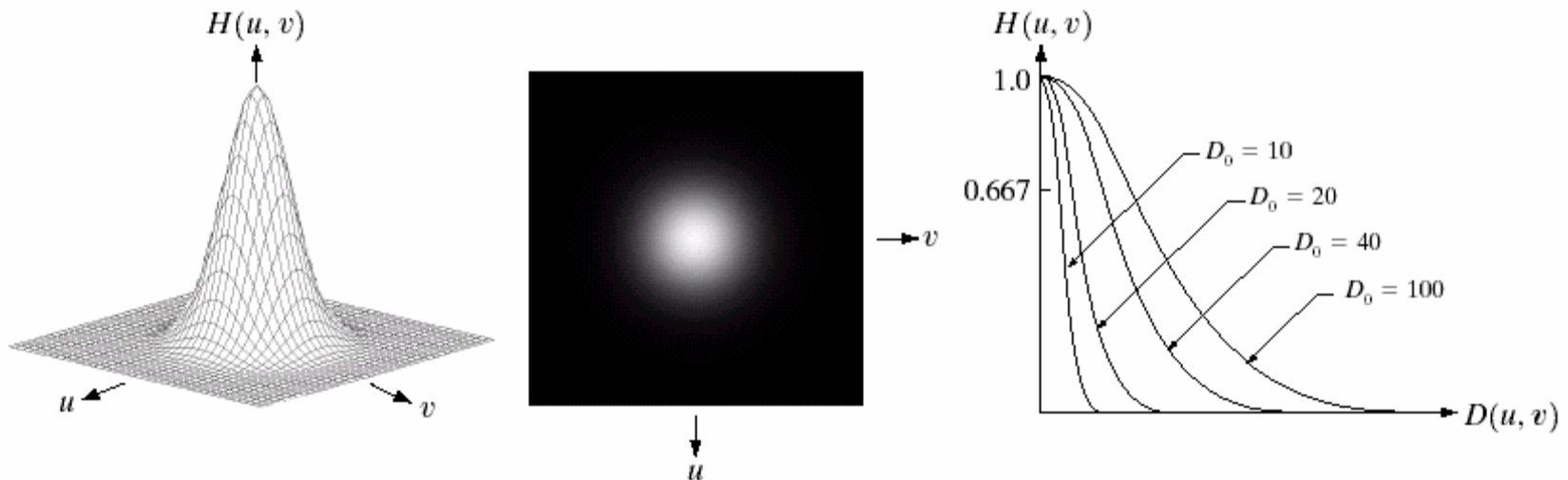


a
b
c
d
e
f

FIGURE 4.15 (a) Original image. (b)–(f) Results of filtering with BLPFs of order 2, with cutoff frequencies at radii of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). Compare with Fig. 4.12.

Gaussian Lowpass Filter: GLPF

The Gaussian filter transfer function: $H(u, v) = \exp[-D^2(u, v)/(2D_0^2)]$



a b c

FIGURE 4.17 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0 .

Example GLPF

GLPF

ILPF

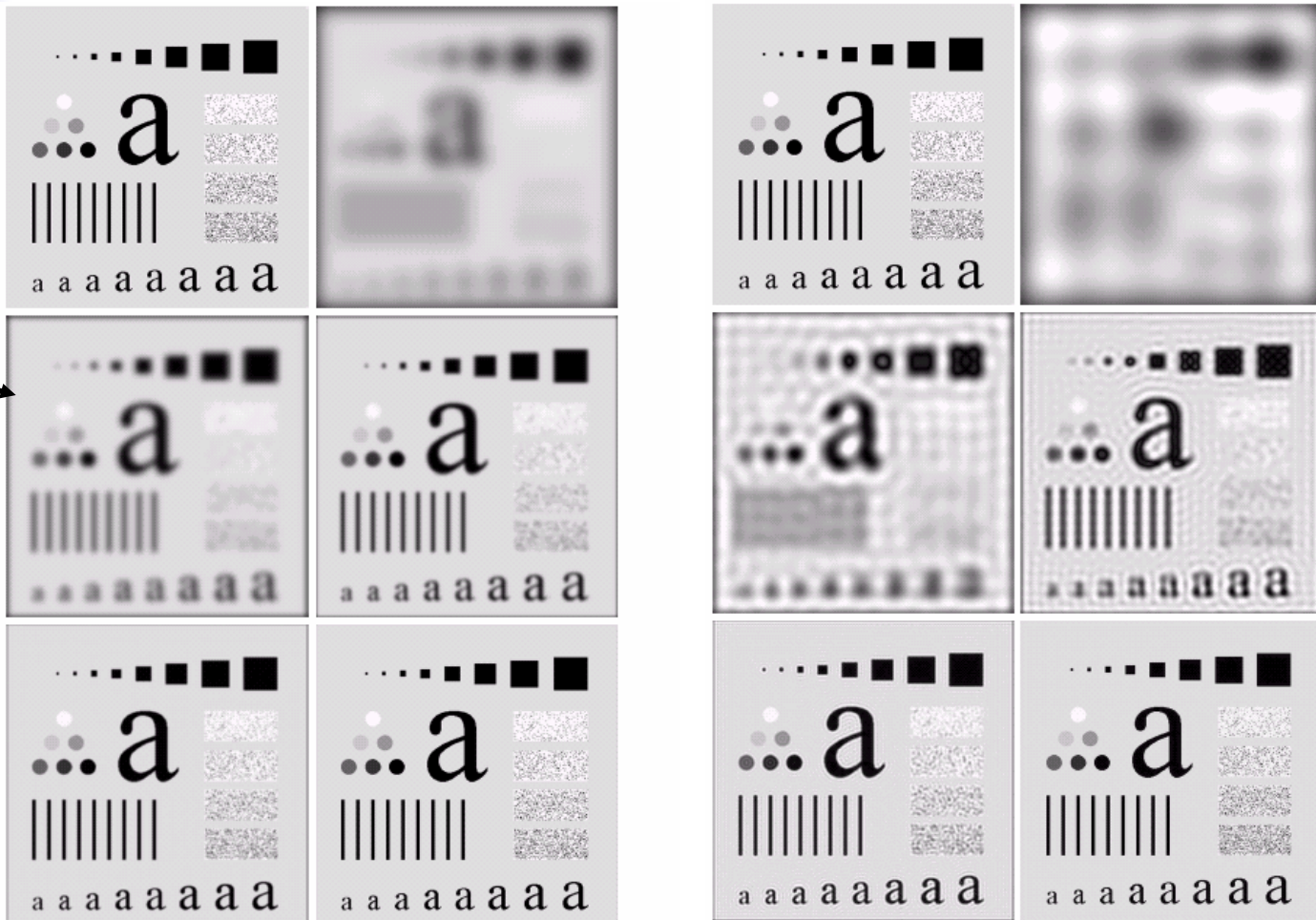


FIGURE 4.18 (a) Original image. (b)–(f) Results of filtering with Gaussian lowpass filters with cutoff frequencies set at radii values of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). Compare with Figs. 4.12 and 4.15.

a b
c d
e f

Example

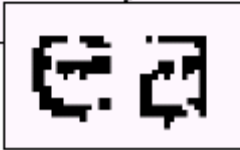
a b

FIGURE 4.19

(a) Sample text of poor resolution (note broken characters in magnified view).

(b) Result of filtering with a GLPF (broken character segments were joined).

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Example



FIGURE 4.20 (a) Original image (1028×732 pixels), (b) Result of filtering with a GLPF with $D_0 = 100$, (c) Result of filtering with a GLPF with $D_0 = 80$. Note reduction in skin fine lines in the magnified sections of (b) and (c).

Sharpening Frequency Domain Filters:

- The Ideal highpass filter

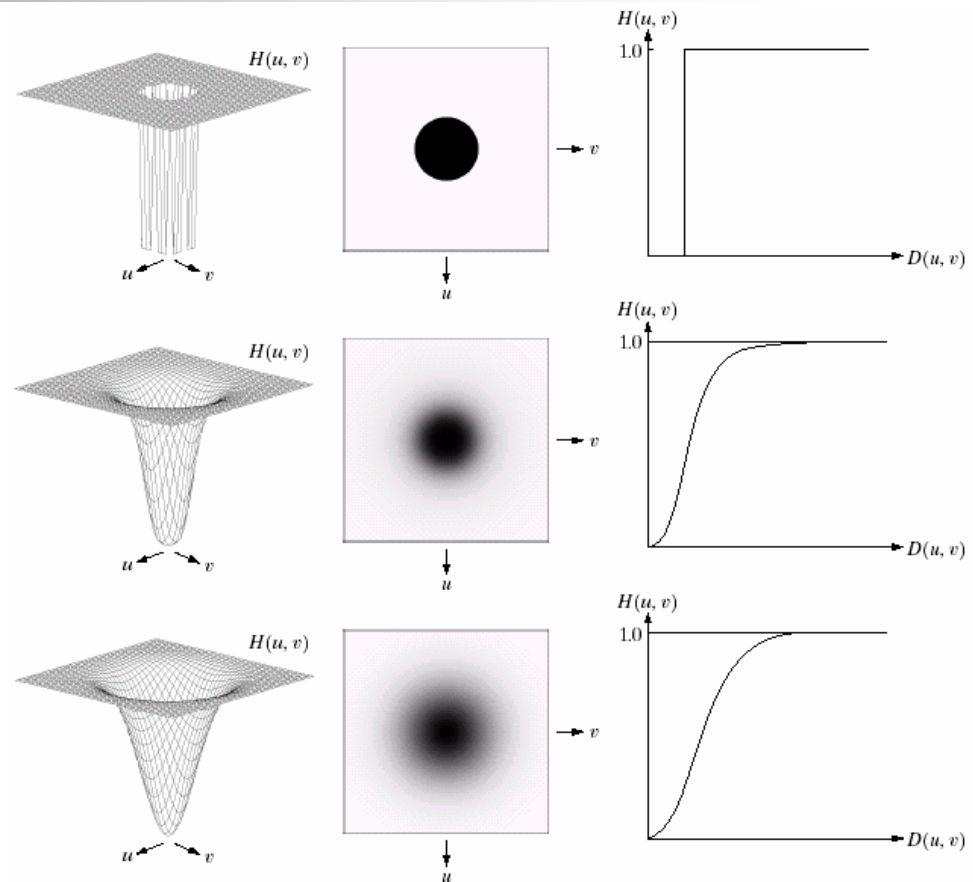
$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$

- The Butterworth highpass filter

$$H(u, v) = \frac{1}{1 + [D_0/D(u, v)]^{2n}}$$

- The Gaussian highpass filter

$$H(u, v) = 1 - e^{-D^2(u, v)/2D_0^2}$$



a b c
d e f
g h i

FIGURE 4.22 Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

Ideal, Butterworth and Gaussian highpass filters: spatial representation

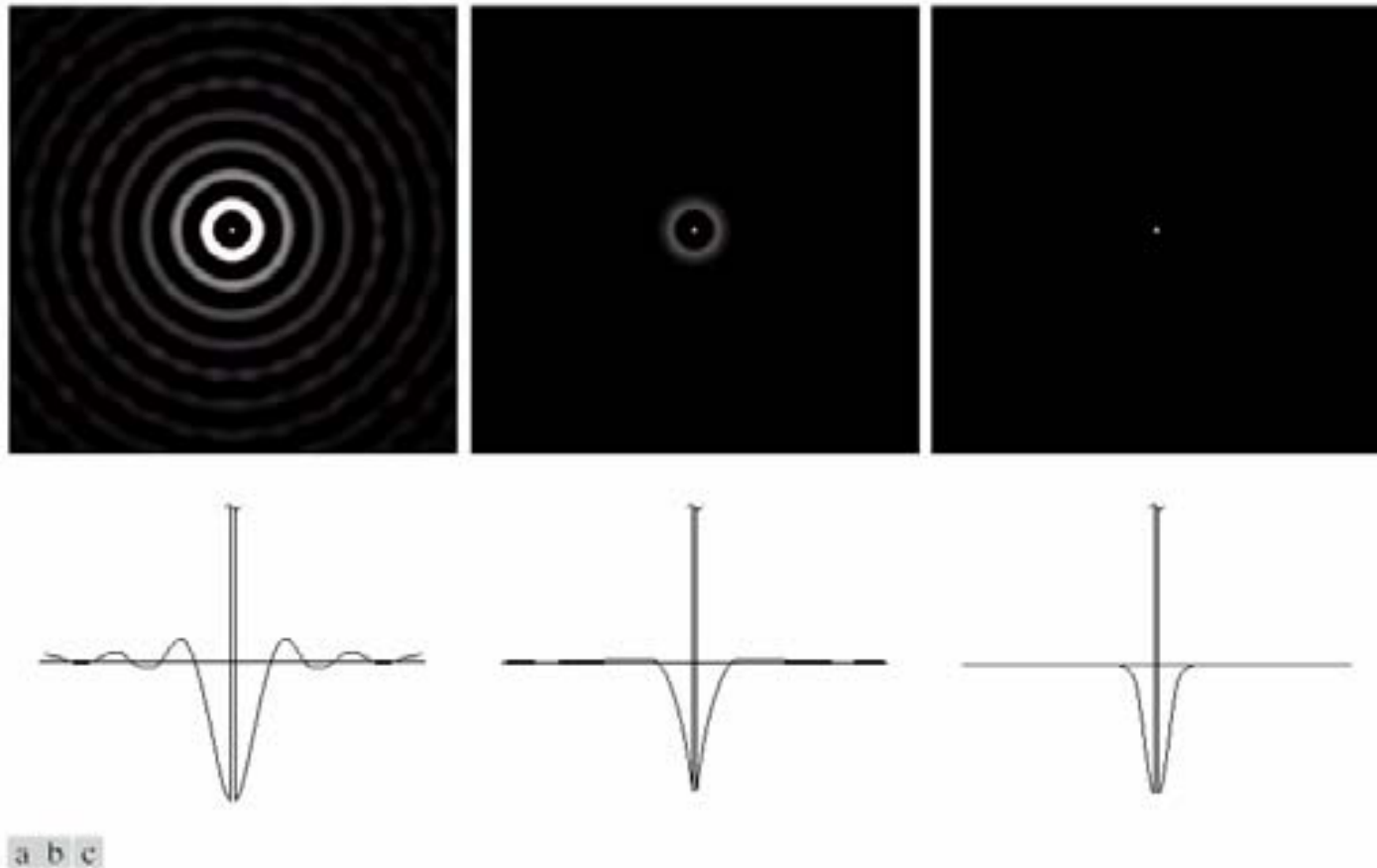
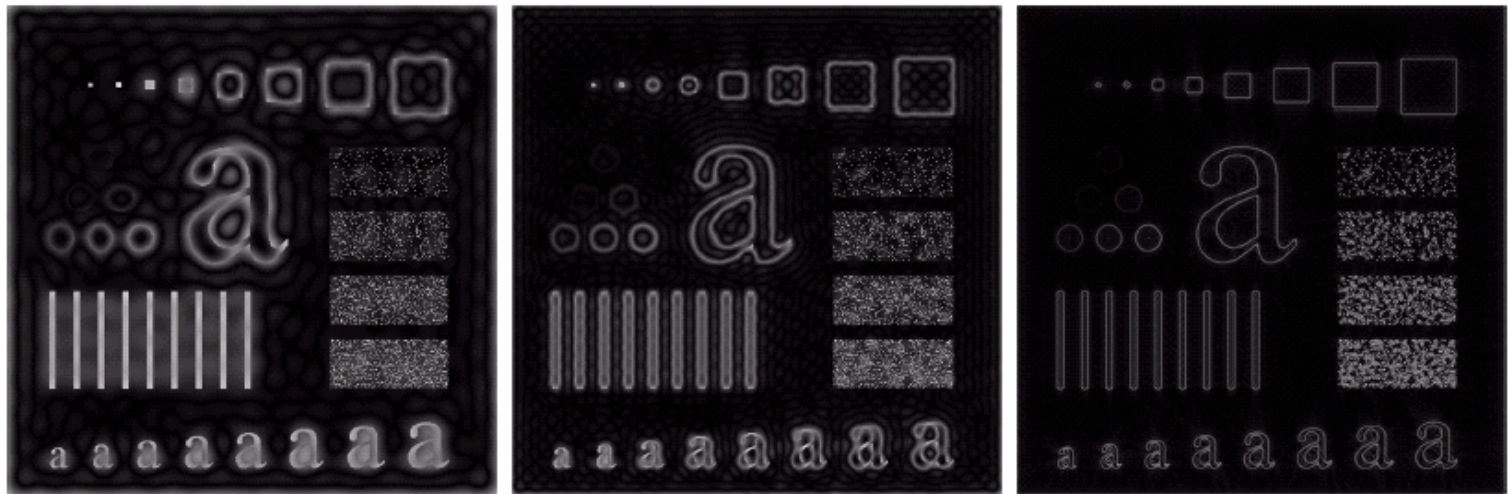


FIGURE 4.23 Spatial representations of typical (a) ideal, (b) Butterworth, and (c) Gaussian frequency domain highpass filters, and corresponding gray-level profiles.

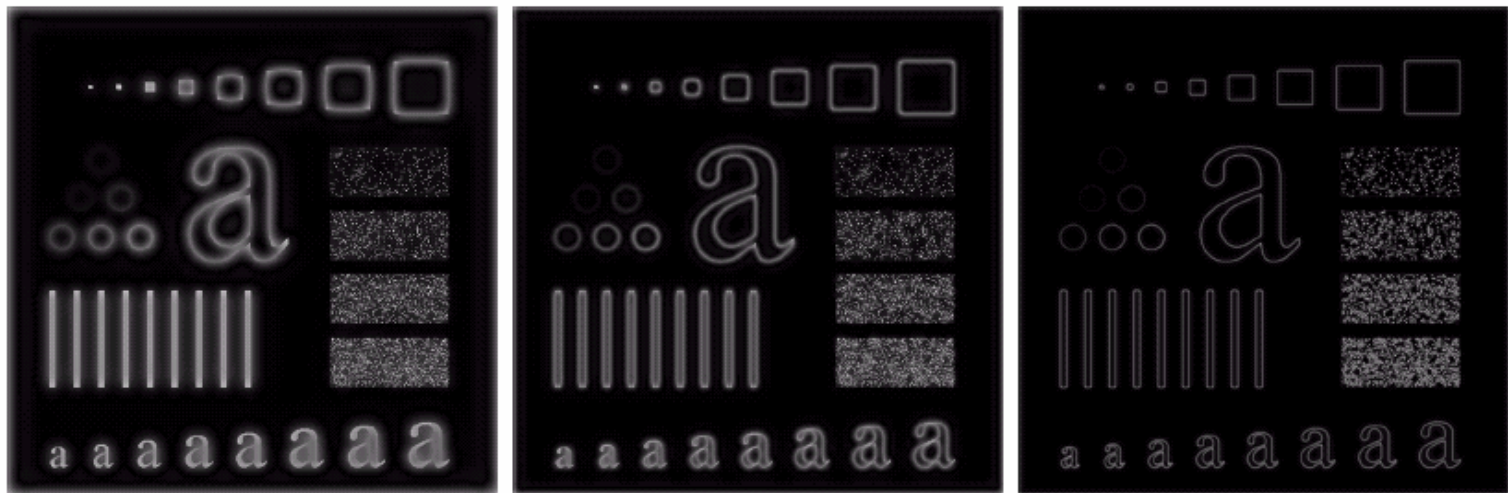
Example: result of IHPF



a b c

FIGURE 4.24 Results of ideal highpass filtering the image in Fig. 4.11(a) with $D_0 = 15$, 30, and 80, respectively. Problems with ringing are quite evident in (a) and (b).

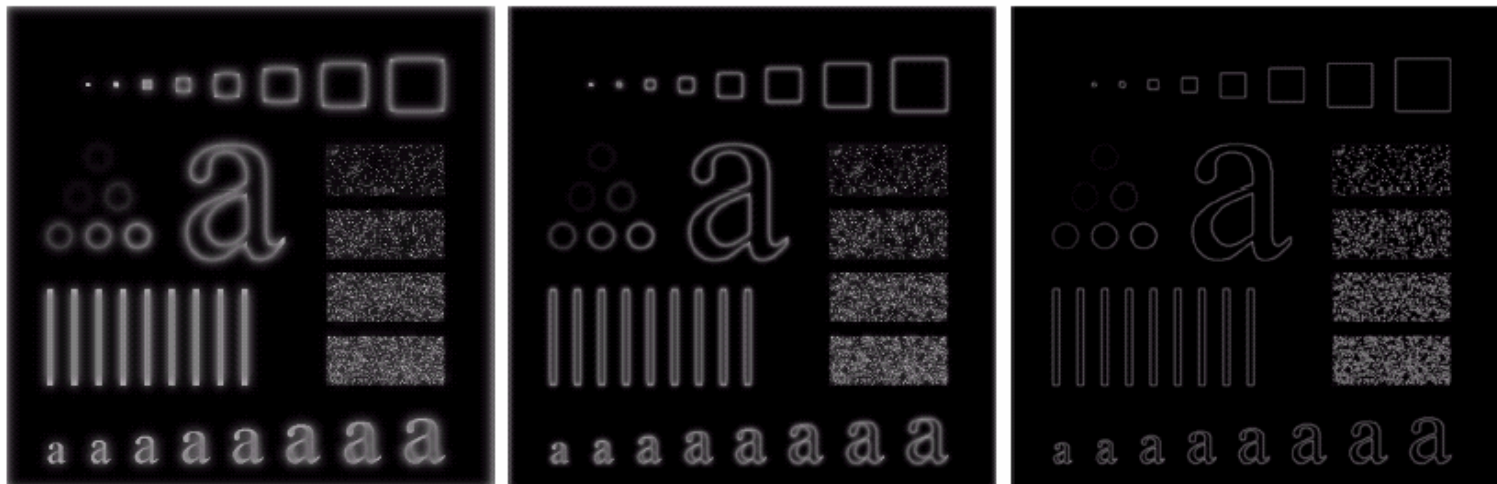
Example: result of BHPF



a b c

FIGURE 4.25 Results of highpass filtering the image in Fig. 4.11(a) using a BHPF of order 2 with $D_0 = 15$, 30, and 80, respectively. These results are much smoother than those obtained with an ILPF.

Example: result of GHPF



a b c

FIGURE 4.26 Results of highpass filtering the image of Fig. 4.11(a) using a GHPF of order 2 with $D_0 = 15$, 30, and 80, respectively. Compare with Figs. 4.24 and 4.25.